

Build a Rock Sorting Key

BUILD • Science • 40 minutes

I can apply a clear rock-grouping rule, check a mystery specimen and defend the placement with evidence.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

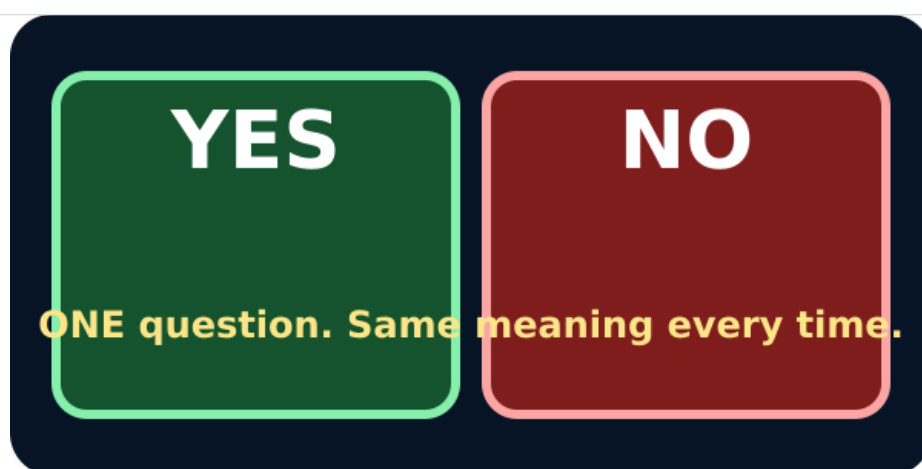
Word	Meaning
criterion	the feature that decides
consistent	same meaning every time
key	questions that sort

First idea

A mystery specimen does not fit the group you expected. What should happen first?

- A. Change the rule only for the mystery so the guess works.
- B. Apply the same stated question to the mystery and accept the evidence-led group.
- C. Hide the mystery because it makes the display harder.

My first idea and reason:



Two mats, YES and NO

We do • choose, explain, check

Code C has faint visible bands. Your rule asks, 'Are visible bands present?' Where should it go?

- A. YES, because faint visible bands still meet the stated rule.
- B. NO, because only very dark bands count today.
- C. Move the boundary after seeing Code C so either answer works.

My choice: _____ Evidence or reason:

Bands-or-Planes Sorting Key

Apply this fixed question to every card: Are visible bands or clear flat sheet-like planes present?

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
YES • BANDS / FLAT PLANES PRESENT	
NO • NOT VISIBLE IN THIS VIEW	

Activity cards

1 • Code A Speckled crystals; no visible bands or flat sheet-like planes.	2 • Code B Fine dark surface; no visible bands or flat sheet-like planes.
3 • Code C Many sand-sized grains and faint but visible bands.	4 • Code D Mostly fine-looking surface; no visible bands or flat planes.
5 • Code E Clear flat, sheet-like planes are visible.	6 • Code F Many large holes; no visible bands or flat planes.

7 • Mystery T

Brown-grey surface with visible bands and no large holes.

Defend Mystery T's placement. Then name another observable rule that would regroup every specimen.

Your lesson record

Create and operate one yes/no sorting key for the coded rock set, place Mystery T and then show how a second rule changes the groups.

Model helps us see

The yes/no mats and visible criterion make consistency easy to check and make any moved boundary obvious.

Model does not show

One two-way appearance key only classifies specimens under its chosen rule. It cannot identify an unknown rock from one image or reveal hidden properties.

Supported

Use one prepared yes/no question, place four reference cards and Mystery T, then say one evidence word for each.

Keep the criterion and matching feature symbol visible. Offer one card at a time and exact adult scribing.

Standard

Write or choose one yes/no criterion, classify all references, place Mystery T and regroup with a second rule.

Use: My criterion is ___. Code ___ goes in YES/NO because ___.

Stretch

Evaluate two different keys for the same set and explain which is more useful for a stated display purpose.

Judge clarity, consistency and what each criterion reveals or hides.

Evidence target

A Sorting Key Case Card with one exact question, all reference placements, defended Mystery T placement and one complete regrouping rule.

Exit, Audience and evidence • W9B

Supported: Complete: My question was ___. Mystery T went in YES/NO because ___.

Standard: Defend Mystery T's placement and explain what changed when the criterion changed.

Stretch: Compare two valid keys and explain why neither key proves Mystery T's rock name.

What I said, and what it changed

Next I will _____.

Adult witness note

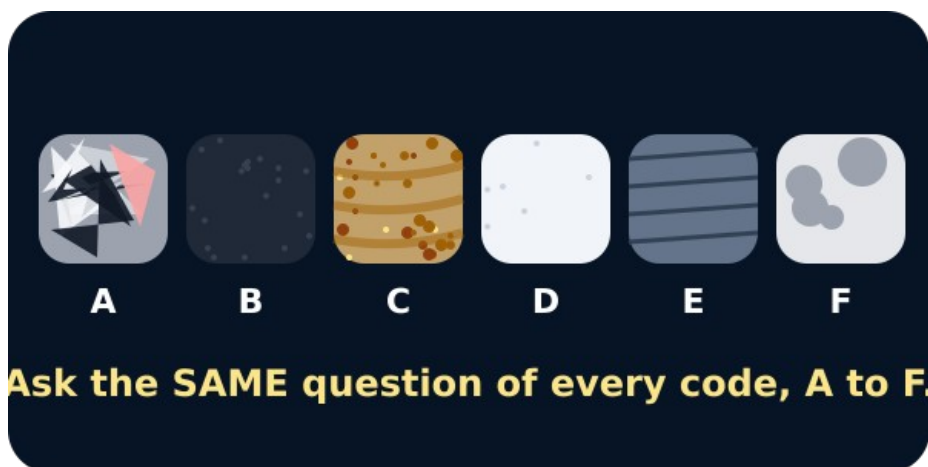
What was observed, and with what level of independence?

My evidence and explanation

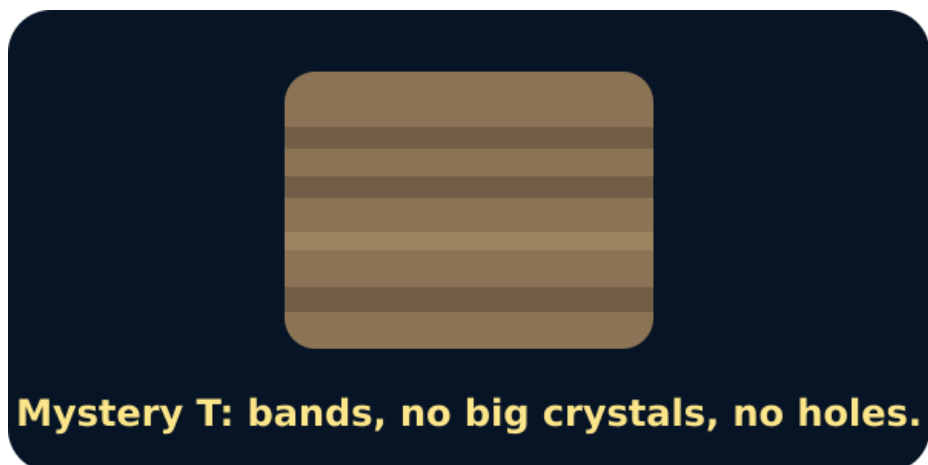
What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



Six coded rocks in a row



The mystery rock T with bands

Exit • one next step

After the adult has listened, what will you keep, improve or check next?